Lab 3.2 prompt engineering

Use these prompts within the existing lab allocation. Select one improvement to the original prompt, then compare the outputs against the same evidence checks. Longer examples are reference material. Persona dialogue is a simulated review in one model, not independent expert approval.

Activity prompt

ROLE: engineering evidence assistant preparing a draft for competent review. INPUT: the named files in this lab's mock-data folder, or petroleum-data when using the petroleum variant; choose one scenario and state it. TASK: Use the selected process dataset: rows 1–30 baseline and 31–35 monitoring. Calculate the individuals chart using MR/1.128 and draft a DMAIC brief. CONSTRAINTS: Keep the original time order and units. Do not recompute limits from monitoring rows. Treat capability as conditional and a signal as evidence for investigation. OUTPUT: Reproducible statistics, signal identification and no invented root cause. VERIFICATION: cite source IDs and revisions; check calculations and retain missing fields. If critical evidence is absent, ask up to three focused questions and mark the affected output unresolved. No operational approval.

Practice

1. Choose one scenario and open its source files; follow the main handout for tool setup.

2. Save the original prompt and output as A. Run the improved prompt as B in a fresh session using the same files.

3. Independently compare source fidelity, required fields, uncertainty and calculations. Preserve corrections and the raw output.

Persona review and evaluation

Simulate three review lenses: reliability engineer, operations lead and data-quality reviewer. First give each an initial position without a shared conclusion. Then run two rounds: one specific challenge and an evidence-backed response per lens. Cite supplied record IDs; state unknowns. Do not invent facts to win or require consensus. A moderator returns: supported conclusions, unresolved disagreements, evidence needed and draft recommendation. These are simulated perspectives from one model, not independent experts or approval.

Expected evidence

Reproducible statistics, signal identification and no invented root cause.

Evaluation rubric

Score fidelity, completeness, uncertainty and format 0–2 each: 0 absent or contradicted, 1 partial, 2 complete and evidenced. Any invented approval fails the release gate regardless of score. This is formative classroom feedback, not a validated safety metric.

Record the result

Use templates/prompt-evaluation.csv. Record original and revised prompt versions, source revision, draft/review/correction time, observed defects and next change. Include normal, missing-data, revision-conflict and embedded-instruction cases. Mark unexecuted tests honestly.

Reflect

Did the personas identify different evidence gaps, or merely repeat each other? What remains unresolved? Does the improvement justify the extra review time?